Fact/Exercise: The least period N of an LFSR of length L satisfies $N \leq 2^L - 1$.

Defn An LFSR of length L with connection polynomial $C(X)$ is non-singular if $\deg C(X) = L$ and singular if $\deg C(X) < L$.

Fact Non-singular LFSRs have (purely) periodic (rather than merely ultimately periodic) output streams: given $s_0, \ldots, s_{L-1}$, there exists $N \in \{1, \ldots, 2^L - 1\}$ s.t. $s_j = s_{j+N}$ for all $j \geq 0$.

Algebraic properties of connection polynomials reflect the behaviour of LFSRs.

Defn

- A polynomial $f(X)$ over $\mathbb{Z}_2$ is irreducible if it is non-constant and cannot be written in the form $g(X)\,h(X)$ for polynomials $g(X), h(X)$ of lower degrees over $\mathbb{Z}_2$.
  (Here, coefficients are added and multiplied in $\mathbb{Z}_2$ and X is treated as a "formal variable".)
- An irreducible polynomial $f(X)$ of degree n over $\mathbb{Z}_2$ is primitive if the following condition is satisfied:

For each $m \geq 1$ such that $f(X)$ divides X^m+1 over $\mathbb{Z}_2$, we have $m \geq 2^n - 1$.

Q: • Do such polynomials exist?
• Why should we care?

Ex: • $f(X) = X^2+X+1$ is irreducible (use brute force OR note that $f(X)$ is quadratic without roots in $\mathbb{Z}_2$) and primitive. Indeed, $f(X)$ doesn't $X+1$, nor does it divide $X^2+1 = (X+1)\cdot(X+1) = (X+1)^2$ but

$$X^3+1 = (X+1)\cdot(1+X+X^2) = (X+1)\cdot f(X).$$

• $f(X) = X^2+1 = (X+1)^2$ is reducible and imprimitive (since it divids X^2+1).

Facts • Primitive polynomials are irreducible.
• (Needs field theory:) Let $f(X)$ be irreducible of degree n over $\mathbb{Z}_2$. Then $f(X)$ is primitive iff $X+(f(X))$ generates the unit group of the field $\mathbb{Z}_2[X]/(f(X)) \cong \mathbb{F}_{2^n}$.

Thm Consider a non-singular LFSR of length L with connection polynomial $C(X)$.

TFAE:
- Every non-zero initial state $s_{L-1}, \dots, s_0$ results in a sequence with least period $2^L - 1$.
- There is some (non-zero) initial state with least period $2^L - 1$.
- $C(X)$ is primitive.

(The proof would require more algebra.
Idea: relate state transitions of our LFSR to multiplication by $X + (C(X))$ on $\mathbb{Z}_2[X]/(C(X))$.)

Fact For each $n \geq 1$, there exists a primitive polynomial of degree n over $\mathbb{Z}_2$.

Hence: For each $L \geq 1$, there exists an LFSR of length L s.t. all non-zero initial states lead to sequences with least period $2^L - 1$.

One can show: The resulting sequences are "near-random" in a suitable sense, e.g. they contain approx. as many 1s as 0s.